

Problem 1

Solution:

(a) By the corollary to axiom 5 on pg 58, a set S containing a single point is considered a rectangle with $h = k = 0$. Therefore, by axiom 5 $S \in \mathcal{M}$ and $a(S) = hk = 0$.

(b) Let n be the total number of points, S_k be the set containing the k -th point, and $S = S_1 \cup S_2 \cup \dots \cup S_n$. That is, S contains all points. Also, if $i \neq j$ then $S_i \cap S_j = \emptyset$ because each point is contained in exactly one S_k . That is, all S_k are mutually exclusive.

We will now show via induction on n that $a(S) = 0$. The base case has already been covered by part (a). For the inductive hypothesis, assume that for arbitrary fixed $1 \leq q < n$ where $S = S_1 \cup S_2 \cup \dots \cup S_q$, $a(S) = 0$.

Now, for the inductive step, consider $S = S_1 \cup S_2 \cup \dots \cup S_n$.

$$\begin{aligned}
a(S) &= a\left(\bigcup_{k=1}^n S_k\right) \\
&= a\left(\left(\bigcup_{k=1}^{n-1} S_k\right) \cup S_n\right) \\
&= a\left(\bigcup_{k=1}^{n-1} S_k\right) + a(S_n) - a\left(\left(\bigcup_{k=1}^{n-1} S_k\right) \cap S_n\right) && \text{(by additive axiom)} \\
&= 0 + a(S_n) - a\left(\left(\bigcup_{k=1}^{n-1} S_k\right) \cap S_n\right) && \text{(by inductive hypothesis)} \\
&= 0 + 0 - a\left(\left(\bigcup_{k=1}^{n-1} S_k\right) \cap S_n\right) && \text{(by part (a))} \\
&= -a\left(\bigcup_{k=1}^{n-1} S_k \cap S_n\right) \\
&= -a\left(\bigcup_{k=1}^{n-1} \emptyset\right) \\
&= -a(\emptyset) \\
&= 0 && \text{(by corollary on pg 58)}
\end{aligned}$$

(c) Lemma: a set L' containing a single line segment is measurable and has zero area.

Proof: By the corollary to axiom 5 on pg 58, a set L' containing a single line segment is considered a rectangle with either $h = 0$ or $k = 0$. Therefore, by axiom 5 $L' \in \mathcal{M}$ and $a(L') = hk = 0$.

From here, you can replicate the approach in part (b). Proof is omitted.

□

Problem 2

Solution:

Lemma I: For any right triangle A we can attach A to a congruent right triangle A' such that A and A' collectively form a rectangle R ($A \cup A' = R$) without overlapping ($A \cap A' = \emptyset$) and A, R share a common length of base and height.

Proof: Omitted. Taken "for free" from Euclidean geometry

Lemma II: Right triangle area is $\frac{1}{2}ab$ where a is length of triangle's base and b is the height.

Proof: Let A be an arbitrary right triangle. By lemma I, there is a corresponding A' such that A and A' collectively form a rectangle. Let R be this rectangle and h, k be the lengths of its sides.

Since A and A' are both right triangles, they are both measurable (by the lemma given in the problem statement). Also, since A and A' are congruent, by the invariance axiom, $a(A) = a(A')$. Further, by axiom 5, $R \in \mathcal{M}$ and $a(R) = hk$. Now, since $A \in \mathcal{M}$ and $A' \in \mathcal{M}$, by the additive axiom we have $a(A \cup A') = a(A) + a(A') - a(A \cap A')$. By substitution, this simplifies to $a(R) = hk = a(A) + a(A) - a(\emptyset) = 2a(A)$ (note: $a(\emptyset) = 0$ by the corollary on pg 58). Thus, $a(A) = \frac{1}{2}hk$. This completes the proof because A, R both share the length of base h and height k (or vice versa).

Lemma III: In any triangle T , we can drop a perpendicular from a vertex to an opposing side. This perpendicular partitions the triangle into two non-overlapping right triangles T_1, T_2 ($T_1 \cup T_2 = T$, $T_1 \cap T_2 = \emptyset$) where s (the length of base of T) = s_1 (length of base of T_1) plus s_2 (length of base of T_2) and T, T_1, T_2 share a common height h .

Proof: Omitted. Taken "for free" from Euclidean geometry

Consider an arbitrary triangle T and drop a perpendicular from any of its vertices to the opposing side s . Lemma III applies. Now from lemma II, observe that $a(T_1) = \frac{1}{2}s_1h$ and $a(T_2) = \frac{1}{2}s_2h$. Also, by the additive axiom, $a(T_1 \cup T_2) = a(T_1) + a(T_2) - a(T_1 \cap T_2)$. By substitution, this simplifies to

$$a(T) = a(T_1) + a(T_2) - a(\emptyset) = \frac{1}{2}s_1h + \frac{1}{2}s_2h + 0 = \frac{1}{2}h(s_1 + s_2) = \frac{1}{2}hs$$

which completes the proof that an area of a triangle is one half its base length times its height. Note: $a(\emptyset) = 0$ by the corollary on pg 58.

□

Solutions are omitted for problems 3 to 6.